Hannah John

Software & Robotics Engineering Student

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SUMMARY

Final-year Software & Robotics Engineering student at QUT with hands-on experience developing PLC software, industrial control systems and Modbus communications. Currently contributing to the development of a commercial battery management system using an Omron PLC. Interested in software engineering, robotics and mechatronics, and enjoy rock climbing outside of work and study.

Experience

CHARG Energy - Student Engineer (BMS Development) 05/2026 - Present

- Developed PLC software for a commercial BMS using Omron CX-Programmer.
- Led the initial BMS prototype, coordinating PLC software development and system integration.
- Designed ladder logic for pumps, sensors and safety systems.
- Implemented reliable Modbus TCP communication between the PLC and external devices.
- Tested and commissioned PLC hardware, sensors and field devices.
- Produced I/O lists, state diagrams and technical documentation.
- Used Git version control within a collaborative engineering team.

Kathron Dry Cleaners – Administration & Operations 06/2021 - 04/2026

- Resolved customer enquiries and supported efficient day-to-day store operations.
- Organised records and improved administrative workflow accuracy.

Finlaysons Timber & Hardware – Dispatch Administration 08/2023 - 03/2024

- Developed a dispatch tracking system for deliveries, suppliers and scheduling.
- Managed invoices, customer orders and supplier communication.

Education

Bachelor of Engineering (Software and Robotics Engineering) Expected 06/2027

Queensland University of Technology

Technical Skills

Industrial and Embedded:

- Omron PLCs
- CX-Programmer
- Modbus TCP
- Industrial Control Systems
- Embedded Systems

Programming Languages:

- JS, Python, HTML/CSS, SQL, C/C++,

Engineering Tools:

- Git, AutoCAD, GitHub, VS code, Linux VM, MySQL, Cursor

Projects

Rental Management Web Application (React, REST API) | Individual Project Semester 1, 2026

- Built a full-stack rental management application using React and REST APIs.
- Developed user authentication, property management and responsive interfaces.
- Designed database interactions and backend API integration.

Email Classification Using Machine Learning | Individual Project Semester 1, 2026

- Developed a transformer-based model to classify emails by author.
- Preprocessed datasets, trained models and evaluated classification performance.
- Improved model accuracy through feature engineering and experimentation.

UAV Dashboard (JavaScript, REST API) | Team Project Semester 2, 2025

- Developed a real-time dashboard displaying live and historical UAV sensor data.
- Connected frontend components to backend APIs and implemented live graph functionality.
- Assisted with sensor integration and system troubleshooting to display live data from 8 environmental sensors.

Embedded Systems for Electric Vehicle | Team Project Semester 1, 2025

- Developed motor control functionality for a microcontroller-based embedded system.
- Assisted with RPM visualisation, real-time graphing, and user interaction features.
- Debugged motor timing & RPM interaction within limited embedded system environments

Interests

Rock climbing • Piano • Robotics • Personal software projects